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RESEARCH ARTICLE



Surface sterilization and moist storage conditions for recalcitrant seeds of three Asian tree species

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ABSTRACT

Ex situ storage of recalcitrant seeds is challenging, due to seed-desiccation sensitivity. Various moist storage techniques have been used to store seeds, but moist conditions also promote fungal infection. This study determined suitable sterilizing agents for surface cleaning and suitable moist storage conditions for three tree species: *Heynea trijuga* Roxb. ex Sims, *Neocinnamomum caudatum* (Nees) Merr., and *Phoebe lanceolata* (Nees) Nees. We used a factorial design with three factors: sterilizing agents, storage temperature and storage media. There were four surface-sterilization treatments: no sterilization (control), 70% ethanol, 3% NaOCl, and 25% metalaxyl. Seeds were stored with two storage media (moist sand and moist filter paper) and at two storage temperatures (room temperature and 4°C). Percentage fungal infection and seed germination were recorded. For *H. trijuga* and *P. lanceolata*, 3% NaOCl (five-minute soaking) was the best sterilizing agent, whilst *N. caudatum* had no optimal sterilizing agent. We recommend storing seeds with moist storage media at 4°C. The best storage media for *H. trijuga* was moist filter paper, while for *P. lanceolata* either moist sand or filter paper was suitable. However, *N. caudatum* seeds should be sown immediately after collection. Our findings emphasize the importance of testing individual species to determine optimal techniques for surface sterilization and storage.

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Introduction

Storing seeds in seed banks is a safe and relatively inexpensive method for long-term *ex situ* conservation of plant genetic material (Hong and Ellis 1996). It is also a backup for *in situ* conservation (Williams 2008). In order to maintain viability, seeds stored in a seed bank must tolerate dry conditions and low temperatures (Pritchard *et al.* 2004). However, not all species produce seeds that can be stored under dry conditions. Seed storage behavior can be divided into three categories based on tolerance to desiccation and low temperature: (1) orthodox seeds (desiccation tolerant seeds) can be dried to low moisture content and stored at low temperatures (-18°C to -25°C) without damage; (2) recalcitrant seeds (desiccation sensitive seeds) do not tolerate desiccation and lose viability, as the moisture content decreases and (3) intermediate seeds can be partially dried but cannot survive low temperatures (Roberts 1973; Ellis *et al.* 1990).

Of the three seed storage behaviors, recalcitrant seeds are the most challenging for storage (Wen 2009). Seed scientists have developed several techniques to store recalcitrant seeds, including partially dry storage,

moist storage, cryopreservation, and storage at the lowest possible temperature without damage, but each of these techniques has its own limitations (Chin 1989; Hong and Ellis 1996; Trusiak *et al.* 2023). The efficiency of recalcitrant seed storage depends on both the technique and seed characteristics of each species (Bonner 2008). Moist storage is an efficient technique for short-term storage of recalcitrant seeds (Kainer *et al.* 1999). Recalcitrant seeds must be stored in moist media to avoid dehydration and at a low temperature to prevent germination but warm enough to avoid chilling injury (Corbineau 2024). Moist media used to maintain moisture include moist sand, sawdust, coconut dust, or perlite (Hong *et al.* 1998). Moist storage must maintain seed viability without promoting seed germination in storage containers. Unfortunately, the moist conditions, necessary to maintain seed viability, sometimes encourages growth of fungi (Calistru *et al.* 2000).

When seeds are collected from the forest floor, they are often contaminated with micro-organisms, particularly fungi (Sutherland *et al.* 2002). Fungal infection reduces seed quality and longevity (Martin *et al.* 2022). Hence, to prolong seed viability, it is important to sterilize the seed surface, before storage (Ramandi *et al.*

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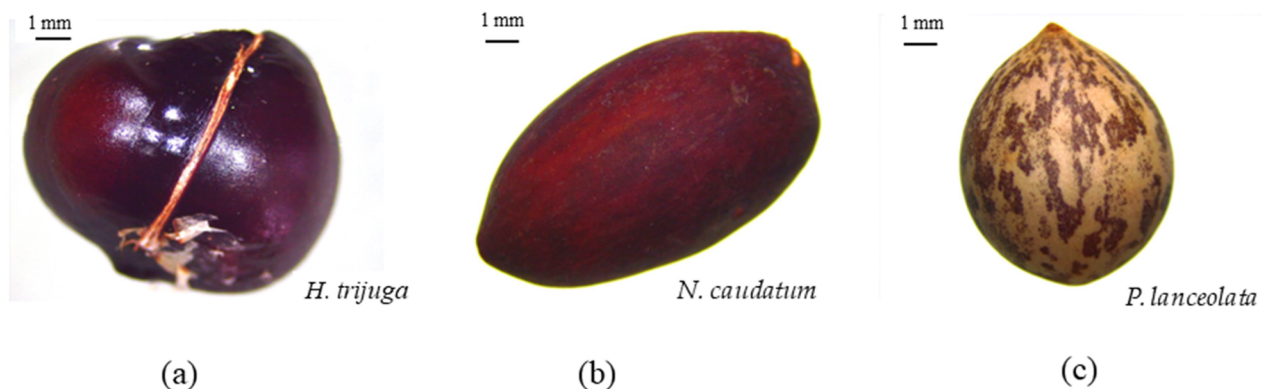


Figure 1. Seeds of three species: a) *Heynea trijuga* b) *Neocinnamomum caudatum* c) *Phoebe lanceolata*.

Table 1. Seed characteristics and species distribution of the three tree species studied. Twenty seeds were used for the measurement of seed mass, length, and width (mean $\pm$ SD).

Species	Family	Seed color	Seed mass (g)	Seed length (mm)	Seed width (mm)	Species distribution ^a
<i>Heynea trijuga</i>	Meliaceae	Blackish purple	0.48 $\pm$ 0.11	13.51 $\pm$ 0.72	9.77 $\pm$ 0.67	550-1,550 m. Bangladesh, Bhutan, Cambodia, South-Central China, Southeast China, India, Indonesia, Laos, Malaysia, Myanmar, Nepal, Philippines, Thailand, Vietnam
<i>Neocinnamomum caudatum</i>	Lauraceae	Reddish brown	0.40 $\pm$ 0.07	15.98 $\pm$ 1.52	7.74 $\pm$ 0.52	900-2,600 m. Bhutan, South-Central China, Southeast China, Northeast India, Laos, Myanmar, Nepal, Thailand, Vietnam
<i>Phoebe lanceolata</i>	Lauraceae	Brown	0.20 $\pm$ 0.03	9.27 $\pm$ 0.36	7.12 $\pm$ 0.34	550-1,600 m. Bangladesh, Bhutan, Cambodia, South-Central China, India, Indonesia, Laos, Malaysia, Myanmar, Nepal, Thailand, Vietnam

^aPOWO (2024).

2019). Commonly used surface-sterilizing agents for plant and seed materials are sodium hypochlorite, ethanol, calcium hypochlorite, mercuric chloride, silver nitrate and hydrogen peroxide (Barampuram et al. 2014), as well as fungicides, such as thiram, carbendazim, captan, mancozeb and metalaxyl (Parisi et al. 2016; Ergin et al. 2021).

Unsuitable storage methods and seed surface sterilization are often linked to low seed germination, seed deterioration and loss of viability. Proper storage conditions may effectively maintain seed viability during storage (Dadjo et al. 2019). Therefore, the objective of this study was to determine effective seed-surface sterilizing agents (chosen among ethanol, sodium hypochlorite and metalaxyl) and suitable moist storage methods (storage media and temperature) for three forest tree species with recalcitrant seeds. The ultimate aim was to support practitioners of forest-ecosystem restoration with sound practicable advice to increase the efficiency of planting stock production of 3 difficult-to-raise recalcitrant tree species.

Materials and methods

Studied species and seed collection

The studied species, *Heynea trijuga* Roxb. ex Sims (Meliaceae), *Neocinnamomum caudatum* (Nees) Merr. (Lauraceae) and *Phoebe lanceolata* (Nees) Nees (Lauraceae), are native to evergreen forests of northern Thailand and are often planted for forest ecosystem restoration (FORRU 2006) (Figure 1, Table 1, see desiccation sensitivity information in Table S1). Fruits and

seeds of the three species were collected from September 2021 to March 2022 from forest restoration plots above Ban Mae Sa Mai, Chiang Mai Province, Thailand (18° 51' 34.71" N 98° 50' 51.28" E, at 1,296 m elevation). Fruits were collected when they were observed to be ripe (on the basis of color) and dropping from the trees i.e. the time of natural seed dispersal—the condition in which restoration practitioners would collect them. For each species, seeds were collected from three different trees. Only intact fruits and/or seeds without signs of fungal infection, animal teeth marks and small holes, caused by insects, were collected. Before seed collection, the trees were checked regularly, and the seeds were collected when they reached maturity, by shaking the trees until the seeds dropped (Berjak et al. 2004). Within three days, the seeds were washed and quality tested, using the floatation technique, to remove empty and damaged seeds, following the protocols of the Forest Restoration Research Unit (Elliott et al. 2013). The seeds were then air-dried for two hours before sorting them for the experiments. Seed handling and experiments were performed at the tree nursery of Doi Suthep Nature Center, Chiang Mai University, Chiang Mai province, Thailand (18° 48' 36.46" N 98° 56' 54.66" E, at 341 m elevation).

Seed moisture content

The initial moisture content (MC) of seeds was measured following the International Seed Testing Association protocol (ISTA 2007). Twenty fresh seeds were randomly selected and weighed. The seeds were

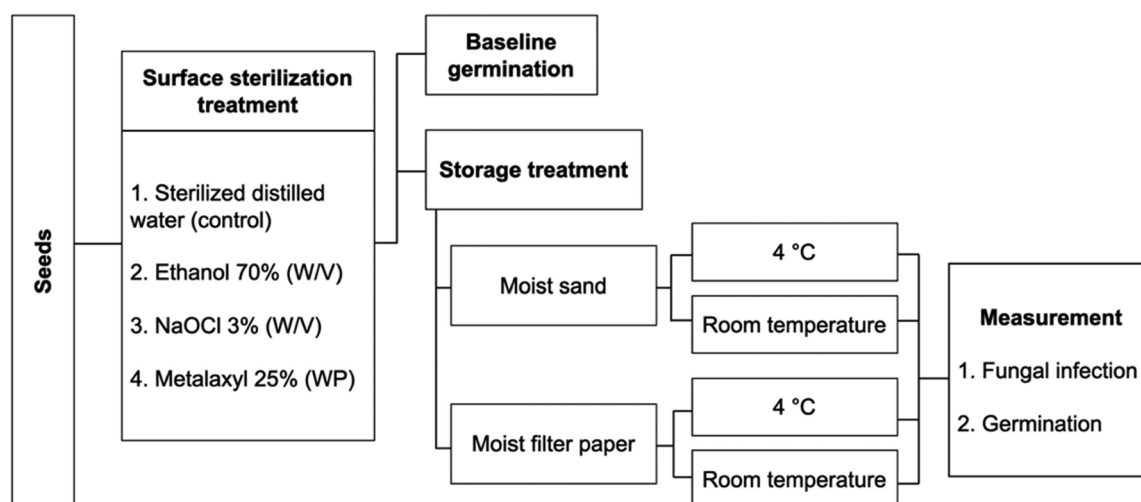


Figure 2. A simplified outline of the experimental protocol to determine the most efficient sterilizing agent and the suitable moist storage conditions for recalcitrant seeds.

then dried in a hot air oven at $103 \pm 3^\circ\text{C}$ for 17h, cooled to ambient temperature by placing them in a container with desiccant beads and reweighed. Moisture content was calculated and expressed as a percentage of fresh weight using the following equation:

$$\% \text{MC} = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Wet weight}} \times 100 \quad (1)$$

Surface sterilization and storage experiments

After washing and air-drying the seeds, the seeds of each species were separated into two groups. The first group was used for a baseline germination test (pre-storage). The second group was used for storage experiments. For each species, there were 16 treatment combinations, consisting of 4 sterilizing agents for surface sterilization $\times$ 2 storage media $\times$ 2 storage temperatures (Figure 2). A hundred seeds were used for each treatment combination.

For seed surface sterilization, the seeds were soaked for five minutes in one of the testing agents: sterilized distilled water (control), ethanol 70% (W/V) (Younesikelaki et al. 2016), sodium hypochlorite (NaOCl) 3% (W/V) (Ramandi et al. 2019), and metalaxyl 25% (WP) (Agrios 2005). As a safety precaution, surface sterilization was conducted in a laminar airflow chamber. The four sterilizing agents were also applied to the group of seeds in the baseline germination test. After soaking in the sterilizing agents, the seeds in each treatment were rinsed three times in sterilized distilled water and dried in an oven at room temperature with a fan on for three hours. For the baseline germination test, the cleaned seeds from each surface sterilizing treatment were separated into three replicates of 30 seeds and sown at a depth of about 1 cm in modular plastic trays, filled with soil.

To identify fast and cost-effective methods for storing seeds after collection for practical purposes, we used room temperature (on average $28.0\text{--}30.5^\circ\text{C}$) and 4°C (fridge) as storage temperature and moist filter

paper and moist sand as storage media. These temperatures and media were chosen because they represent the temperature conditions available in most nurseries. The cleaned seeds were stored in air-tight glass containers with either moist sand or moist filter paper and stored either at 4°C or room temperature. Before storing seeds, the storage containers and equipment were autoclaved at 15 psi pressure at 121°C for 15 min followed by cooling to room temperature. In the case of the moist sand treatment, the sand was dried overnight in a hot air oven at 120°C , followed by cooling to room temperature before use. For moist sand preparation, we filled a rectangular container (1,650 ml) with heat-sterilized sand to a depth of 3 cm and added sterilized distilled water to the sand at a ratio of 1:2 by volume (water to sand). To avoid the sand being excessively wet, we tilted the container and let the excessive water drain from the sand until the water stopped flowing out. A layer of seeds was placed on the moist sand, and then another 1 cm layer of moist sand (prepared in the same way) was added on top. For the treatment with moist filter paper, two sheets of filter paper were saturated with sterilized distilled water and placed in a rectangular container (1,650 ml). To avoid direct contact with the moist filter paper, the seeds were put in a Petri dish and the Petri dish was placed in the same container next to the moist filter paper. The storage containers were closed and placed in either a climate chamber (MEMMERT-ICH 110L) at 4°C or under ambient room temperature on a lab bench. The average room temperature in this study was between 28.0 to 30.5°C .

After 30 days of storage, the seeds were inspected for surface fungal infection under a stereo microscope. For each treatment, we recorded the number of fungal infected seeds based on visible signs of fungal infection, i.e. hyphae, mycelium, and/or spores. For the seed germinability test, for each treatment three replicates of 30 seeds each were sown at a depth of about 1 cm in modular plastic trays with soil in a tree nursery. The number of germinated seeds were recorded weekly over three months.

Data analyses

The R programming language (version 4.2.2) (R Core Team 2021) with RStudio (version 2022.07.2) (RStudio Team 2020) was used for all analyses. Seed fungal infection and germination were analyzed separately for each species. We used a generalized linear model (GLM) to analyze our data. The independent variables were the sterilizing agent, storage media and storage temperature. The dependent variable was either the probability of seed fungal infection or the probability of germination. For fungal infection, a binomial distribution was used as the error distribution. For germination data of all species, a quasi-binomial family was used because of overdispersion (the ratio of residual deviance to degrees of freedom >1). The model started with all independent variables included in the model (full model). Variables were then eliminated one at a time (a stepwise algorithm with step function in R) until the model with the lowest Akaike Information Criteria (AIC) was obtained. Furthermore, if differences among treatment levels were observed, we used Tukey's HSD tests to determine which treatments differed from one another. In addition, for the fungal infection data, we used a t-test to determine whether the differences between means of the two storage media treatments and the two temperature treatments were significant for each species.

Results and discussion

Seed moisture content

H. trijuga seeds had the highest initial moisture content (MC) at $51.2 \pm 3.1\%$, followed by *P. lanceolata* ($33.8 \pm 1.4\%$) and *N. caudatum* ($14.9 \pm 5.0\%$).

Seed fungal infection

For *H. trijuga*, the GLM indicated that sterilizing agents, storage temperature and storage media affected fungal infection ($P < 0.01$). There were two-way interaction effects between storage temperature and sterilizing agents ($P < 0.01$). For seeds cleaned with sterilized distilled water (control), the percentage of seeds with fungal infection was high for both storage temperatures (approx. 42–87%). For seeds cleaned with 3% sodium hypochlorite (NaOCl), the percent fungal infection was much lower for seeds stored at 4°C (approx. 1%) in comparison to those stored at room temperature (approx. 96%). Moreover, the effect of storage media depended on the temperature at which seeds were stored ($P < 0.05$). When stored with moist filter paper, percent fungal infection did not differ significantly between the two storage temperatures. When storing seeds in moist sand, the percent fungal infection at room temperature was higher than at 4°C (Figure 3). Of the four sterilizing agent treatments, 3% NaOCl was the most effective at lowering fungal infection for *H. trijuga* ($47.8 \pm 27.4\%$ averaged across treatments) (Table 2). Considering storage temperature, the t-test indicated that fungal infection was significantly lower when seeds were stored at 4°C (Table 2).

For *N. caudatum*, the GLM indicated that sterilizing agents, temperature and storage media significantly influenced the percent fungal infection ($P < 0.01$). There were two-way interaction effects between storage media and sterilizing agents. When stored with moist filter paper, the percent fungal infection did not differ among sterilizing agents (approx. 96–100%), while the percent fungal infection was the lowest when the seeds were cleaned with 70% ethanol and stored in moist sand (approx. 3%). The t-test showed significant differences among storage media (Table 2). Storing seeds with moist filter paper resulted in a higher fungal infection compared with moist sand (Figure 3).

For *P. lanceolata*, the GLM indicated that sterilizing agents, temperatures and storage media significantly influenced the percent seed fungal infection ($P < 0.01$). There were two-way interaction effects between storage temperature and sterilizing agents ($P < 0.01$). For seeds cleaned with sterilized distilled water (control), 70% ethanol, and 25% metalaxyl, percent fungal infection was high at both storage temperatures (approx. 40–60%). For seeds cleaned with 3% NaOCl, fungal infection was much lower at 4°C (approx. 3%) compared with room temperature (approx. 43%). The effect of storage media depended on the temperature at which seeds were stored ($P < 0.01$). At 4°C, fungal infection did not differ significantly between the storage media, while at room temperature it was higher when the seeds were stored with moist filter paper (Figure 3, 4). Of the four sterilizing agent treatments, 3% NaOCl was the most effective at lowering fungal infection for *P. lanceolata* ($23.0 \pm 14.7\%$ averaged across treatments) (Coefficient estimate $\pm$ SE = -4.41 ± 0.502 , $z = -8.790$, $P < 0.01$) (Table 2). The t-test indicated that the percent fungal infection was significantly lower when the seeds were stored in moist sand than with moist filter paper (Table 2).

Our findings suggest that seed different tree species require different agents to achieve effective surface sterilization. NaOCl was the most effective surface sterilizing agent for two of the three species tested (*H. trijuga* and *P. lanceolata*). Ramandi et al. (2019) and Younesikelaki et al. (2016) reported similar results. Ramandi et al. (2019) found that soaking seeds with 3% NaOCl for five minutes was the best method for seed surface sterilization of *Catharanthus roseus* (L.) G. Don (Apocynaceae) and that 3% NaOCl did not negatively affect seed germination. Furthermore, Younesikelaki et al. (2016) reported that a suitable sterilization technique for *Althaea officinalis* L. (Malvaceae) was to clean with 4% NaOCl for five minutes. In contrast, we found that seed fungal infection in *N. caudatum* was high and did not differ among sterilizing agents. For *N. caudatum*, 70% ethanol seemed to be a promising sterilizing agent for lowering fungal infection, but the concentration and duration of seed soaking require further investigation. The appropriate concentration and duration of treatments depend on the size and structure of the seeds (Cetinkaya et al. 2022).

Seeds stored in moist sand had a lower fungal infection percentage than those stored with moist filter

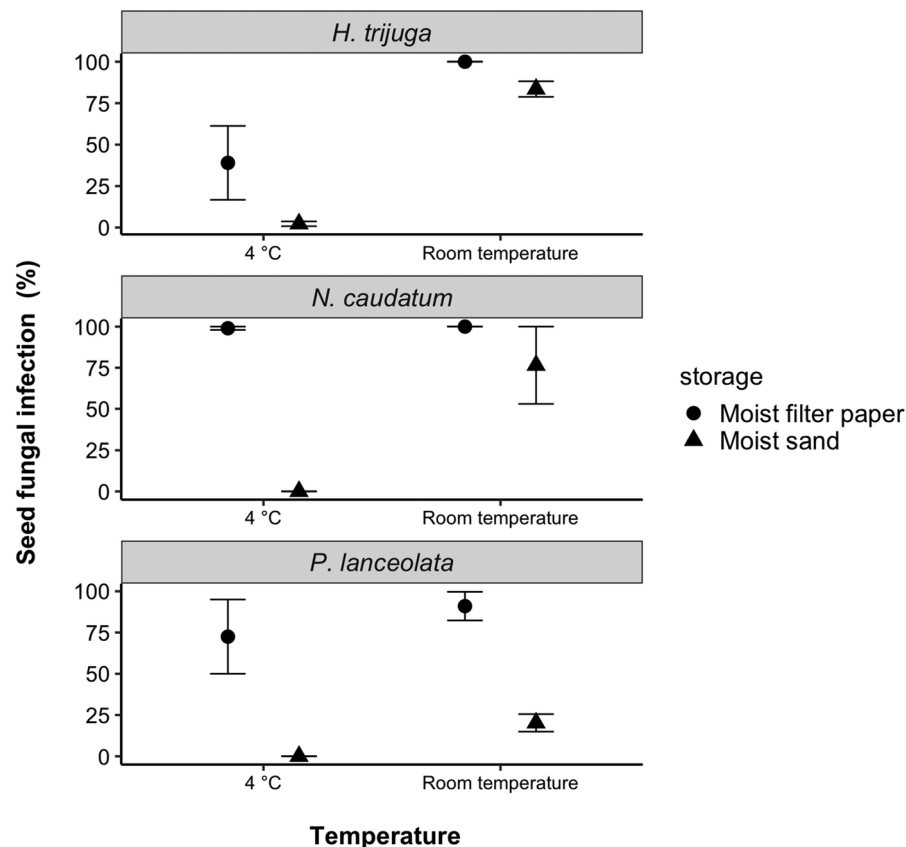


Figure 3. Seed fungal infection of three native tree species in different storage treatments.

Table 2. The percentage of seed fungal infection (mean $\pm$ SE) among three factors for the three studied species. Within each factor of interest, values sharing the same letters ('a' or 'b') are not significantly different ($P < 0.05$).

Species	Fungal infection (%)							
	Sterilizing agent				Storage media		Temperatures	
	Control	Ethanol 70%	Metalaxyl 25%	NaOCl 3%	Moist filter paper	Moist sand	4 °C	Room temp.
<i>H. trijuga</i>	64.5 $\pm$ 21.2 ^a	48.3 $\pm$ 27.9 ^{ab}	64.3 $\pm$ 20.3 ^a	47.8 $\pm$ 27.4 ^b	69.5 $\pm$ 15.5 ^a	42.9 $\pm$ 15.5 ^a	20.6 $\pm$ 12.4 ^b	91.8 $\pm$ 3.8 ^a
<i>N. caudatum</i>	75.0 $\pm$ 25.0 ^a	51.5 $\pm$ 28.0 ^a	75.0 $\pm$ 25.0 ^a	74.0 $\pm$ 24.7 ^a	99.5 $\pm$ 0.5 ^a	38.3 $\pm$ 18.1 ^b	49.5 $\pm$ 18.7 ^a	88.3 $\pm$ 11.8 ^a
<i>P. lanceolata</i>	51.3 $\pm$ 28.2 ^a	53.0 $\pm$ 23.5 ^a	56.5 $\pm$ 25.4 ^a	23.0 $\pm$ 14.7 ^b	81.8 $\pm$ 11.7 ^a	10.1 $\pm$ 4.5 ^b	36.3 $\pm$ 17.2 ^a	55.6 $\pm$ 14.2 ^a

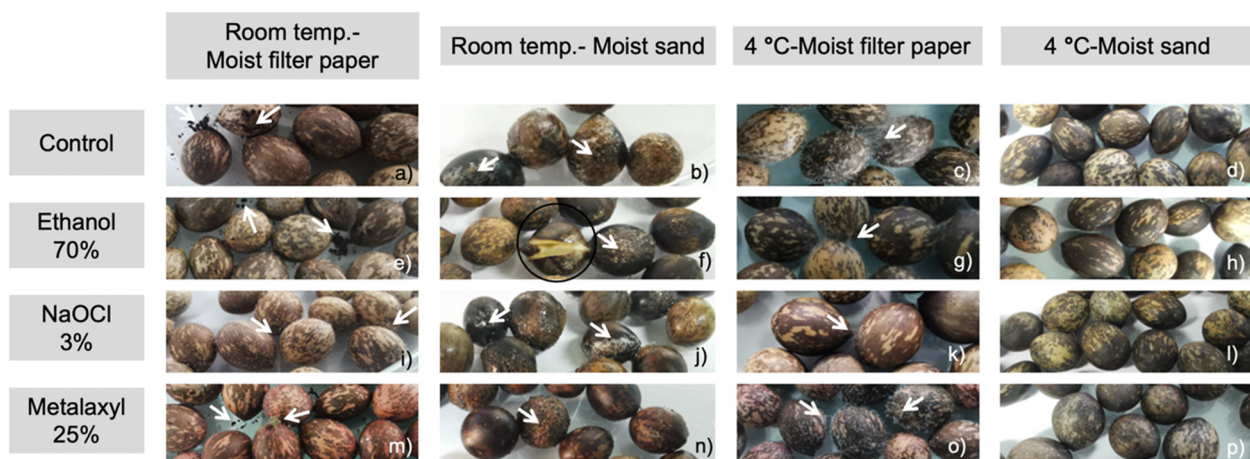


Figure 4. Examples of visible seed fungal infection under different treatments of *Phoebe lanceolata* seeds. Arrows indicate fungal infection, and the circle in (f) shows a seed germinating during storage.

paper (Table 2, Figure 3). The infectivity of fungi varies with environmental conditions (Mensah-Attipoe and Toyinbo 2019). Infection may originate from spores present in seeds before storage or from spores

already in storage containers (Martin et al. 2022). When storing seeds in moist filter paper, they are in direct contact with the air inside the container. The spores may become airborne and circulate within the

Table 3. Percent seed germination (mean $\pm$ SE) compared among three factors for the three studied species. Within each factor of interest, values sharing the same superscripts ('a' or 'b') are not significantly different ($P < 0.05$).

Species	Germination (%)									
	Sterilizing agent				Storage media			Temperatures		
	Control	Ethanol 70%	Metalaxyl 25%	NaOCl 3%	Baseline	Moist filter paper	Moist sand	Baseline	4 °C	Room temp.
<i>H. trijuga</i>	15.8 $\pm$ 4.5 ^a	15.8 $\pm$ 6.0 ^a	14.0 $\pm$ 4.8 ^a	16.4 $\pm$ 5.9 ^a	30.8 $\pm$ 7.4 ^a	16.0 $\pm$ 4.0 ^a	7.4 $\pm$ 2.5 ^b	30.8 $\pm$ 7.4 ^a	23.1 $\pm$ 3.5 ^a	0.3 $\pm$ 0.3 ^b
<i>N. caudatum</i>	6.9 $\pm$ 5.1 ^a	1.1 $\pm$ 0.8 ^a	10.0 $\pm$ 4.7 ^a	1.6 $\pm$ 1.1 ^a	13.6 $\pm$ 6.6 ^a	0.0 $\pm$ 0.0 ^a	5.4 $\pm$ 2.8 ^a	13.6 $\pm$ 6.6 ^a	5.4 $\pm$ 2.8 ^a	0.0 $\pm$ 0.0 ^a
<i>P. lanceolata</i>	61.6 $\pm$ 4.7 ^a	53.3 $\pm$ 4.1 ^a	57.6 $\pm$ 5.6 ^a	55.6 $\pm$ 6.9 ^a	58.1 $\pm$ 6.3 ^a	58.9 $\pm$ 4.5 ^a	54.6 $\pm$ 4.0 ^a	58.1 $\pm$ 6.3 ^a	42.5 $\pm$ 2.9 ^b	71.0 $\pm$ 3.3 ^a

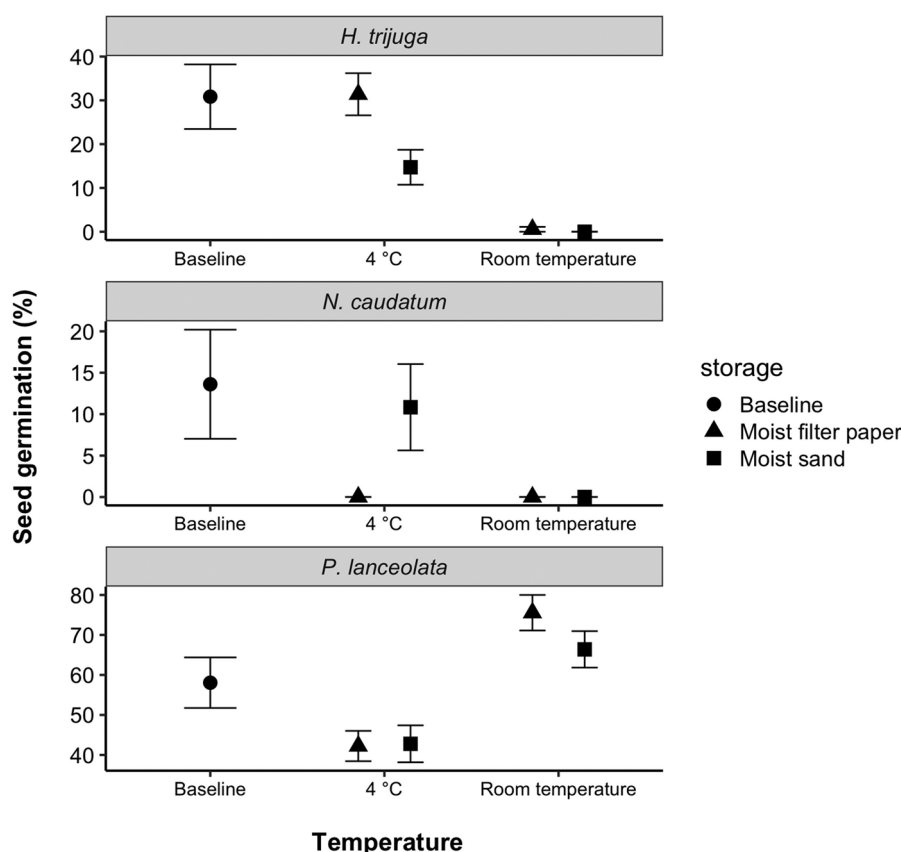


Figure 5. Seed germination of three native tree species in different storage treatments.

container, thus increasing infection. When storing seeds in moist sand, the seeds were covered in sand. The sand may provide a barrier to air-borne transmission of fungal spores and lower the chance of infection. Therefore, storing seeds in moist sand may yield a lower percentage of fungal infection than with moist filter paper, since with the latter seeds are not covered.

Seed germination

P. lanceolata had the highest percent baseline (pre-storage) seed germination at 58%, followed by *H. trijuga* (31%), and *N. caudatum* (14%) (Table 3). One month of moist storage did not significantly reduce the percent germination of *N. caudatum* and *P. lanceolata* (Table 3). For all species, the sterilizing agents did not significantly reduce seed germination compared to the control treatment (Table 3). This finding indicates that the concentration of sterilizing agents did not negatively affect seed germinability.

Different species responded differently to storage conditions. For *H. trijuga*, the GLM indicated

significant differences in the percent germination among storage media and storage temperatures ($P < 0.05$). There were no three-way or two-way interactions between the factors. After storing the seeds in containers in moist sand, percent germination decreased significantly by approximately 23% (Coefficient estimate $\pm$ SE = -0.994 ± 0.233 , $z = -4.269$, $P < 0.01$) (Table 3, Figure 5). Storing seeds at room temperature significantly decreased the percent seed germination of *H. trijuga* by approximately 30% (Coefficient estimate $\pm$ SE = -4.73 ± 0.392 , $z = -5.3$, $P = 0.0014$) (Table 3, Figure 5). *H. trijuga* seeds had the highest seed moisture content (MC). We found that the seeds became rotten during storage in moist sand.

For *N. caudatum*, the sterilizing agents, storage temperatures, and storage media did not significantly affect seed germination. The seed germination of *N. caudatum* was low to begin with (14%) (Figure 5). We observed that storing seeds for one month with moist filter paper and at room temperature resulted in no seed germination. For *N. caudatum*, moist storage is not recommended and seeds should be sown right away after collection (Table 3, 4).

Table 4. Recommendation for each species.

Species	Seed surface sterilization	Moist storage conditions
<i>H. trijuga</i>	Soak in 3% NaOCl for five minutes, then rinse with water.	Seeds should be stored with moist filter paper at 4°C (in moist sand, the seeds rotted and had low germination).
<i>N. caudatum</i>	None (further investigation is needed to test different sterilizing treatments).	No moist storage. Seeds should be sown immediately after collection. If seeds must be stored, store them in moist sand at 4°C for no more than one month.
<i>P. lanceolata</i>	Soak in 3% NaOCl for five minutes, then rinse with water.	Seeds can be stored either with moist filter paper or in moist sand at 4°C.

In contrast, storing *P. lanceolata* seeds at room temperature resulted in the highest germination (Table 3, Figure 5)—13% significantly higher than at 4°C (Coefficient estimate $\pm$ SE = 1.196 ± 0.213 , $z = 5.631$, $P < 0.01$). *P. lanceolata* seeds began to germinate in the storage container under moist conditions at room temperature (Figure 4f). Ma et al. (2023) reported that metabolic and biochemical activities in the seeds stored under moist conditions or warm stratification at 28–30°C may break seed dormancy. Furthermore, Malngiang et al. (2023) reported that warm stratification ($25 \pm 1^\circ\text{C}$) and moist sand storage for 30 days broke seed dormancy of *Ilex embelioides* Hook.f. (Aquifoliaceae).

In general, we do not recommend using room temperature for storing seeds (Table 3, 4). However, if breaking dormancy is a goal, exposing seeds to moist conditions at room temperature for short periods will often promote seed germination.

Our study suggests that storing recalcitrant seeds at low temperatures maintains seed viability in the short term. Low temperatures limit the metabolic processes necessary for germination and development by slowing food mobilization and reducing enzyme activity. Conversely, higher temperatures cause plants to have a faster metabolism, losing the seed energy needed for growth (Haj Sghaier et al., 2022). This finding agrees with those of previous studies of many plant species. Nagendra et al. (2019) reported that the recalcitrant seeds of *Syzygium cumini* (L.) Skeels (Myrtaceae) can be stored for 13 days at 5°C without significant viability loss. Wang et al. (2021) reported that moist seed stratification in sand at 4°C for 100 days is the best method for storing *Manglietia crassipes* (Y.W.Law) V.S.Kumar (Magnoliaceae). The results of Subhashini Devi et al. (2012) suggest that storing the *Sterculia urens* Roxb. (Malvaceae) seeds at 0–4°C (refrigerator) in a polythene bag retained the best germination percentage and seed vigor index when compared with seeds stored at room temperature ($30 \pm 2^\circ\text{C}$). For recalcitrant seeds, it is best to sow the seeds immediately after collection. However, if sowing immediately is impossible, moist storing recalcitrant seeds at a low temperature (4–5°C) is preferable to leaving the seeds to dry.

Conclusions

With an aim towards developing techniques to facilitate seed conservation of forest tree species, this study determined an efficient sterilizing agent for seed surface sterilization and suitable moist storage conditions for three recalcitrant seeded species. Our findings

demonstrate that surface sterilization and a low temperature storage (4°C) are beneficial for the short-term storage of recalcitrant seeds by reducing fungal infection. Moreover, storing recalcitrant seeds in moist conditions is necessary to prevent dehydration. However, no one technique fits all species. Therefore, to better handle recalcitrant seeds after seed collection, individual species should be tested for surface sterilization and optimal conditions for seed storage.

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Author contributions

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Disclosure statement

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Data availability statements

The raw data supporting the conclusions of this article will be made available by the authors on request.

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